MATHEMATICS — FINAL EXAM

MARKING SCHEME — teacher copy, not for distribution

Grade 11AG · 26/10/2026 · Total: 100 marks · Anvarbek Khaydarov

Award method marks for a correct rule stated and applied, even where the final answer is wrong.

SECTION A (6 questions × 4 marks = 24 marks)

1. Find $\lim_{x\to 2}(3x^2 - 5)$. [4 marks]
Answer: 7
2. Find $\lim_{x\to 3}(x^2 - 9)/(x - 3)$. [4 marks]
Answer: 6
3. Find $\lim_{x\to\infty}(5x^2 + 1)/(2x^2 - x)$. [4 marks]
Answer: 5/2
4. Differentiate $y = 4x^5$. [4 marks]
Answer: $20x^4$
5. Differentiate $y = \sin x + e^x$. [4 marks]
Answer: $\cos x + e^x$
6. Differentiate $y = \ln x - \cos x$. [4 marks]
Answer: $1/x + \sin x$

SECTION B (6 questions × 6 marks = 36 marks)

7. Find $\lim_{x\to 1}(x^2 + 2x - 3)/(x - 1)$. [6 marks]
Answer: 4
8. Find $\lim_{x\to\infty}(3x^3 - 2x)/(6x^3 + 5)$. [6 marks]
Answer: 1/2
9. Differentiate $y = (2x + 1)^5$. [6 marks]
Answer: $10(2x + 1)^4$
10. Differentiate $y = x^2e^x$, giving your answer in factorised form. [6 marks]
Answer: $xe^x(x + 2)$
11. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. [6 marks]
Answer: $4\sec^2 4x$; gradient 4
12. Differentiate $y = \ln(3x^2 + 1)$. [6 marks]
Answer: $6x/(3x^2 + 1)$

SECTION C (3 questions, 40 marks)

13. Evaluate the following limits:
(a) $\lim_{x\to 0}(\sqrt{(x + 4)} - 2)/x$ [6 marks]

Answer: 1/4

(b) $\lim_{x\to\infty} (\sqrt{(x^2 + 3x)} - x)$ [7 marks]

Answer: 3/2

14. Differentiate, simplifying your answer where possible:

(a) $y = \sqrt{(x^2 + 9)}$ [6 marks]

Answer: $x/\sqrt{(x^2 + 9)}$

(b) $y = \ln(\cos x)$ [7 marks]

Answer: $-\tan x$

15. A curve has equation $y = x^2 \sin x$.

(a) Find dy/dx. [7 marks]

Answer: $2x \sin x + x^2 \cos x$

(b) Find d^2y/dx^2. [7 marks]

Answer: $2 \sin x + 4x \cos x - x^2 \sin x$